ASSESSMENT TOOL 1

ANALYTICAL PROBLEM SOLVING

Course Name	Cloud Computing and Big Data Analytics
Course Code	CSA15
Course Outcome	CO2 – Analyze virtualization architectures to identify performance and security issues in cloud computing environments (BL3)
Dave's Taxonomy	Manipulation (Level 2)
Question Count	5
Total Marks	50
Student Name	Paleru Dharani Govardhan
Register Number	192525280
Programme / Branch	B.Tech – Artificial Intelligence and Machine Learning

CODE OF CONDUCT

I, **Paleru Dharani Govardhan**, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Paleru Dharani Govardhan*

ASSESSMENT TASKS

Analytical Problem Solving

1. Comparison of Type-1 and Type-2 Hypervisors for a Mission-Critical Cloud Datacenter

A hypervisor is a virtualization layer that allows multiple virtual machines (VMs) to share the physical resources of a host. Type-1 and Type-2 hypervisors differ mainly in where this layer runs.

A Type-1 hypervisor, also called a bare-metal hypervisor, runs directly on the physical server hardware. Examples include VMware ESXi, Microsoft Hyper-V and Xen. Since it does not depend on a general-purpose host operating system, it has a shorter software path between the VMs and hardware. This generally provides better performance, lower overhead and more predictable resource management. It also reduces the number of host

operating-system components that could be attacked. Centralized management, resource scheduling, live migration and high-availability features make it suitable for large datacenters.

A Type-2 hypervisor runs as an application on top of a host operating system. Examples include Oracle VirtualBox and VMware Workstation. It is easy to install and useful for development, testing and desktop virtualization. However, VM operations pass through the host operating system, which can add overhead. A failure or compromise of the host OS can also affect the virtual machines.

For a cloud datacenter hosting mission-critical applications, Type-1 is the better choice. It offers stronger operational control, efficient resource utilization and features needed for high availability. The provider should still apply secure configuration, patch management, access controls, network segmentation and continuous monitoring. A Type-2 hypervisor is more appropriate for personal testing or development rather than a large production cloud environment.

2. Workload Isolation and Attack Surfaces in a Shared Cloud Environment

A cloud provider may host finance, healthcare and student portal workloads on the same physical hardware. Virtualization enables this arrangement by dividing a physical server into separate virtual machines, each with allocated CPU, memory, storage and virtual network interfaces. A hypervisor controls access to physical resources and keeps the execution environments of different VMs logically separated.

Isolation can be strengthened through separate virtual networks, VLANs or software-defined network policies, security groups, dedicated storage volumes and role-based access control. Sensitive workloads can be placed in separate resource pools or on dedicated hosts when their risk or compliance requirements demand stronger separation. The provider should enforce least privilege and prevent one tenant from accessing another tenant's VM images, disks, snapshots or management interfaces.

Two major attack surfaces are:

→ **Hypervisor and management plane:** A vulnerability in the hypervisor, virtual device emulation or management interface could allow an attacker to escape a VM or gain control over multiple workloads. Exposed management APIs and weak administrator credentials increase this risk.

→ **Virtual network and inter-VM communication:** Incorrect network rules, insecure virtual switches or poorly isolated traffic can allow unauthorized communication between tenants. Attackers may exploit exposed services or misconfigured network paths to reach another workload.

Other risks include compromised VM images, unpatched guest operating systems and side-channel attacks that infer information from shared hardware resources. To reduce these threats, the provider should patch hypervisors, restrict management access, use multi-factor authentication, scan VM images, encrypt sensitive data, monitor network traffic and regularly test isolation controls. Virtualization provides the foundation for separation, but secure configuration and continuous monitoring are essential to maintain it.

3. Analysis of VM Host Utilization and Corrective Actions

The utilization snapshot indicates that the host is under heavy resource pressure:

• CPU utilization: 92% – This is very high and suggests that the physical processor is close to saturation. Too many active VMs, CPU-intensive applications or inefficient workloads may be competing for processor time.

• Memory utilization: 88% – Most physical memory is already in use. The host may have limited capacity for new VMs. If memory becomes insufficient, swapping or ballooning may occur, which can reduce application performance.

• Storage I/O latency: 35 ms – This value indicates slow storage response for many latency-sensitive workloads. Possible causes include overloaded disks, insufficient IOPS, shared storage contention, slow network-attached storage or backup operations running at the same time.

• Average VM boot time: 170 seconds – This is relatively long and may be caused by CPU and memory contention, slow storage reads, large VM images, startup services or network delays.

The likely overall bottleneck is resource contention across CPU, memory and storage. These problems can affect one another: slow storage can delay VM startup, while high CPU and memory use can make initialization services slower.

Corrective actions should be prioritized as follows:

1. **Identify heavy workloads:** Use hypervisor monitoring dashboards to identify VMs consuming unusually high CPU, memory or I/O. Check workload patterns before changing resource allocations.
2. **Reduce CPU contention:** Move busy VMs to less-loaded hosts, balance workloads across the cluster, and add CPU capacity if demand remains high. Avoid excessive vCPU allocation because it can increase scheduling contention.
3. **Improve memory availability:** Right-size VM memory, remove unused VMs and increase host RAM where necessary. Monitor swapping and ballooning rather than relying only on the utilization percentage.
4. **Resolve storage latency:** Check disk queue depth, IOPS and throughput. Move high-I/O workloads to faster storage such as SSD-backed tiers, distribute workloads across storage pools and schedule backups outside peak periods.
5. **Improve VM startup:** Use optimized VM templates, remove unnecessary startup services and investigate storage and network delays. Keep frequently used images readily available where supported.
6. **Set monitoring alerts:** Establish thresholds for sustained CPU, memory and I/O pressure. Review trends to plan capacity before performance becomes unacceptable.

The organization should measure performance again after each change. Adding hardware without identifying the actual bottleneck may increase cost without solving the underlying issue.

4. Software Defined Datacenter Compared with a Traditional Datacenter

A traditional datacenter depends heavily on dedicated physical hardware and manually configured infrastructure. Servers, storage devices and network equipment are often managed separately. Provisioning a new application environment may require purchasing hardware, configuring devices and coordinating several teams. This can make scaling and changes slow.

A Software Defined Datacenter (SDDC) uses software to abstract and manage compute, storage and networking resources. These resources can be provisioned, configured and monitored through centralized management tools and automation. This improves agility because infrastructure can be created or modified through software policies and APIs rather than requiring manual changes to every physical device.

The main virtualization considerations are:

→ **Compute virtualization:** A hypervisor divides physical servers into multiple VMs. Administrators can create, resize, migrate and manage VMs using centralized tools. This improves hardware utilization and allows workloads to be moved when a host is overloaded or requires maintenance.

→ **Storage virtualization:** Physical storage from multiple devices can be presented as logical storage pools. Software-defined storage can automate allocation, replication, snapshots and policy-based placement. This simplifies capacity management and supports flexible expansion.

→ **Network virtualization:** Software-defined networking (SDN) separates network control from the underlying forwarding hardware. Virtual networks, routing and security policies can be configured programmatically. Network segmentation can also isolate application environments and tenants.

For example, when an organization launches a new application, an SDDC platform can automatically provision VMs or containers, allocate storage and configure network policies using predefined templates. In a traditional datacenter, the same task may require several manual hardware and network configuration steps.

SDDC improves speed, scalability, consistency and resource utilization. However, it also creates a need to secure centralized management systems, automation tools and APIs. Strong authentication, role-based access, change auditing, backups and regular patching are necessary to prevent a management-plane compromise from affecting the entire datacenter.

5. Multi-Cloud Identity Mismatches and Secure Federation Design

A multi-cloud federation allows an organization to use services from more than one cloud provider while enabling coordinated identity and access management. A project may fail when providers use different identity formats, user identifiers, role names, authentication methods or permission models. For example, a user may be identified by an email address in one provider and by a unique subject identifier in another. Directly mapping these values can create duplicate accounts, incorrect permissions or unauthorized access.

The organization should establish a central identity provider (IdP) as the trusted source for user authentication. Cloud providers can be connected to this IdP using standard federation protocols such as Security Assertion Markup Language (SAML 2.0) or OpenID Connect (OIDC). The IdP authenticates the user and issues a signed assertion or token containing verified identity information and relevant claims. Each cloud provider validates the token and maps approved claims to its own roles and permissions.

A secure design should include:

1. **Consistent identity mapping:** Use a stable, unique user identifier rather than relying only on changeable email addresses. Define standard attributes and maintain clear mappings between central groups and provider-specific roles.
2. **Least-privilege authorization:** Grant only the permissions required for each job. Use separate roles for administration, development, finance and read-only access. Avoid long-lived shared credentials.
3. **Strong authentication:** Enforce multi-factor authentication, secure session settings and short-lived tokens. Disable access promptly when a user leaves or changes roles.
4. **Secure token handling:** Validate token signatures, issuer, audience and expiry. Protect signing keys, rotate them periodically and use encrypted communication such as TLS.
5. **Privacy protection:** Share only the identity attributes needed by each cloud service. Avoid sending unnecessary personal or sensitive information. Apply data minimization, encryption and appropriate retention policies.
6. **Auditing and monitoring:** Record authentication events, role changes and privileged actions across providers. Send logs to a protected central monitoring system to detect suspicious activity.

For example, a company may run its customer application on one cloud provider and its analytics platform on another. Employees authenticate through the central IdP. The application cloud receives only the claims needed to operate the application, while the analytics cloud grants access only to approved analyst roles. This federation approach provides consistent authentication and controlled access without creating separate passwords for every cloud. Regular testing of trust relationships and permission mappings is essential to prevent identity-related security failures.

ASSESSMENT RUBRIC

Criteria	Weightage	Level 4 – Exemplary
Problem Interpretation	20%	Interprets all technical requirements accurately.
Analytical Reasoning	25%	Provides strong reasoning with clear cause-and-effect analysis.

Method Selection	20%	Chooses highly appropriate virtualization and security concepts.
Solution Quality	20%	Provides feasible and technically sound solutions.
Presentation of Analysis	15%	Presents analysis in a well-structured and technically precise manner.

TOTAL MARKS AWARDED: _____ / 50